Mira: Product Requirements Document (PRD)

Organization: Nexora Pvt. Ltd. · **Product:** Mira · **Product owner and security / compliance:** Phina Enemuo · **Sponsor:** Interview Kickstart · **Version:** 1.0 (as built, replaces v0.2) · **Related:** Mira BRD v1.0

Sources: field names and expected values come from the four project files (project_description.txt, project_timeline.csv, project_risks.csv, sample_task_board.csv) and the baseline test inputs in eval_mira_inputs.txt. Project dates come from the task board (2026-04-07 to 2026-08-04). The as-of date for time-based tests is 2026-04-20. Version 1.0 adds a Status column (Met, Partly met, Not built) to the requirements and records the results.

1. Product overview

Vision. Give every PM/TPM at Nexora a reliable assistant that turns raw project data into plans, risk assessments, status reports and milestone alerts in minutes, without inventing anything.

Core principle: grounded or silent. Mira only states what the data supports. When data is missing or too thin, it says so and asks for what it needs.

Users. Primary: PMs and TPMs. Secondary: the sponsor, engineering leads and the client, who receive PM-approved outputs.

Built on: n8n Cloud (Mira Chat, Mira Core, Model Call, Mira Eval, Mira Judge Calibration), OpenAI gpt-4o-mini, Google Drive and Sheets, Gmail for approval and send, and Langfuse for traces and scores.

2. Goals and non-goals

Goals: generate plans, risk assessments and weekly status reports in a consistent format. Flag blocked tasks and at-risk milestones. Refuse to fabricate when input is insufficient. Make every run observable and measurable.

Non-goals (v1): live task board API integration, sending to stakeholders without approval, fine-tuning, market analysis, resource allocation, and an audience-tailored stakeholder update generator.

3. User stories

ID	As a...	I want to...	So that...	Status
US-1	PM	give Mira a project description and get a structured plan	I skip 3 to 4 hours of manual planning	Met
US-2	PM	get a risk matrix specific to my project	risks are consistent and relevant	Met
US-3	PM	upload the task board and get a weekly status report	I skip 1 to 2 hours of compiling	Met
US-4	PM	ask "which tasks are blocked or at risk?"	I can act before deadlines slip	Met
US-5	PM	ask "what milestones are coming in the next 2 weeks?"	I can prepare in advance	Met
US-6	PM	get an update email for a given sprint	I can communicate quickly and professionally	Partly met. T12 gives a Sprint 2 update. Audience-tailored versions are not built
US-7	PM	be told when my request lacks detail	I don't receive a made-up plan	Met
US-8	sponsor	see quality, cost and failure data for Mira	I can trust and fund it	Met: Langfuse traces, the results sheet and the failure analysis

4. System design

The architecture diagram is in the Mira build writeup and in the repository as `docs/mira_architecture.png`.

4.1 Orchestration pattern: Router

A request is loaded with its project data, classified once by the orchestrator, and sent to one of five branches: Clarify, Planner, Risk, Status or Milestone. Only Clarify skips human approval, because it generates no project content. The whole run is traced in Langfuse.

Why a Router. In order of weight:

1. **Refusing is a routing decision.** Four baseline tests (T2, T4, T6, T9) pass only if Mira asks for detail and generates nothing. With a router, Clarify is a path of its own, so thin requests never reach a generator that would be tempted to fill the gaps.
2. **The jobs share almost no logic.** A plan, a risk matrix, a status count and a milestone check need different data, templates and checks. Separate branches keep each prompt short and easy to test.
3. **It is cheaper and faster.** One small classifier call plus one generator call, instead of running every agent on every request. The orchestrator call costs about $0.00008 and the average request about $0.00032.
4. **Failures are easy to locate.** Each of the 12 baseline tests maps to one branch, so a failure points to one prompt or one rule.
5. **It fits n8n.** The Switch node does the routing, and each model call goes through one shared Model Call sub-workflow.

Pattern	What it is	Why it was not chosen
Single agent with tools	One prompt handles every request	Five jobs in one prompt. Refusal behavior is hard to guarantee and harder to test
Sequential pipeline	Every request passes through every step	Runs unneeded agents each time, costs more, lets content from one job leak into another
Hierarchical (supervisor and workers)	A supervisor plans, delegates and merges	The tasks are independent, so the supervisor adds calls, latency and one more place to hallucinate
Router (chosen)	Classify once, run one specialist	Matches independent job types and makes refusal a path of its own

Trade-off. The router depends on its classifier. A wrong intent sends the request to the wrong branch, so the classifier returns strict JSON, is tested on all 12 baseline inputs, and sends unclear requests to Clarify.

4.2 Design decisions

1. **Deterministic data layer.** CSV parsing, status counts, sprint filtering, due-date comparison and blocked-task detection run in code. The model receives computed facts and writes the narrative.
2. **Sufficiency check in the router.** Requests with too little detail never reach a generator.
3. **Context stuffing, no RAG.** The project files are small, so they are passed directly in the prompt.
4. **Fixed as-of date.** Every date comparison uses an explicit as-of date, shown on every output.
5. **One small model.** OpenAI gpt-4o-mini runs the orchestrator and all branches. A comparison with gpt-4.1-mini was planned but not run, because gpt-4.1-mini is blocked on the project's OpenAI allow-list (KL-7). OpenAI is the only approved provider.
6. **Human approval by email.** Gmail send-and-wait with an approval form. Clarify replies skip it.

5. Functional requirements and status

5.1 Orchestrator

ID	Requirement	Status
FR-O1	Classify each request into plan, risk, status, milestones or other (clarify)	Met
FR-O2	Decide whether the request has enough detail, and list what is missing for generation intents	Met (T2, T4, T6, T9)
FR-O3	A request is insufficient if there is no project description for plan or risk requests, or no task data for status requests	Met
FR-O4	Return strict JSON. Invalid JSON retries once, then goes to Clarify	Met

ID	Requirement	Status
FR-O5	Requests naming the ABCDE Ltd. project use the provided project files	Met

5.2 Clarify branch

ID	Requirement	Status
FR-C1	Explain briefly why more information is needed	Met
FR-C2	Ask for scope, goals or deliverables, timeline and team size	Met
FR-C3	For status requests without data, say no task data is available and report no progress, percentages or task names	Met (T6)
FR-C4	Never produce a partial plan, risk list or status as a best guess	Met

5.3 Planner

ID	Requirement	Status
FR-P1	Produce phases, key activities, milestones, deliverables and timeline estimates	Met (T1)
FR-P2	Phases and milestones come from the timeline and description. Every goal is traceable to the description	Met in the baseline. One intermittent wrong start date was seen in Run 3 (KL-1)
FR-P3	Mark items not in the source as "Proposed, not in source", or omit them	Partly met. Not separately tested
FR-P4	Follow the standard plan template	Met

5.4 Risk assessor

ID	Requirement	Status
FR-R1	Produce a categorized risk matrix: category, risk, impact, mitigation, source ID	Met (T3)
FR-R2	For "top N risks", rank only risks in the risk CSV and cite each by ID	Met (T7)
FR-R3	Any additional risk must be labelled as an AI suggestion and be relevant to logistics or AI adoption	Partly met. The judge flags risks not in the register (the fake R11 test). Weak justifications still pass (KL-6)
FR-R4	Exclude generic risks unrelated to the project	Met in the baseline

5.5 Status reporter

ID	Requirement	Status
FR-S1	Code computes counts for Done, In Progress, To Do and Blocked and lists task IDs and names	Met (T5, T10)
FR-S2	The report shows tasks by status, blocked items, upcoming milestones and overall health with a justification	Met
FR-S3	Task names match the CSV. No invented tasks, owners or percentages	Met
FR-S4	Sprint-specific requests filter to that sprint only	Met
FR-S5	Show the as-of date on every report	Met

ID	Requirement	Status
FR-S6	On request, format the report as an update email for a named sprint	Partly met. T12 gives a Sprint 2 update without Sprint 3+ tasks. An intermittent "no blocked items" overclaim was seen in Run 3 (KL-2). Audience-tailored versions are not built

5.6 Milestone tracker (extended capability)

ID	Requirement	Status
FR-M1	Code compares milestone dates and task due dates with the as-of date and task status	Met
FR-M2	Flag a task or milestone as At risk when it is blocked, overdue, or due within the alert window with open work	Met. Tasks due soon and in progress go to a Watch list (KL-3)
FR-M3	Each flag has a reason and a suggested action. T024 is identified	Met (T8)
FR-M4	"Upcoming milestones in the next 2 weeks" lists only milestones from the timeline file	Met (T11). Run 2 was a false pass and was fixed in Run 3 (KL-4)

5.7 Observability

ID	Requirement	Status
FR-L1	Each request creates one Langfuse trace	Met. A trace has a root, the orchestrator call and the specialist call
FR-L2	Traces carry metadata: test ID, intent, model, as-of date	Met for model, as-of date and route. Test ID is on baseline runs
FR-L3	Token usage, cost and latency are recorded per call	Met

ID	Requirement	Status
FR-L4	Evaluation scores are attached to traces	Met: the risk_relevance evaluator scores risk answers

5.8 Human review (approval gate)

ID	Requirement	Status
FR-H1	Every generated plan, risk assessment, status report and milestone alert is held until a PM approves it	Met. Clarify replies are exempt
FR-H2	The reviewer can Approve, Reject with a reason, or Edit and approve	Met. The Gmail form has a Decision, a Reason and an Edited text field
FR-H3	A rejection captures a reason category plus free text and can trigger one regeneration with the feedback	Partly met. Free text and edited text are captured. Categories and regeneration are not built
FR-H4	Email send-and-wait approval, with pending items expiring after a timeout and never sent automatically	Met. Pending approvals expire after 24 hours and nothing is sent automatically
FR-H5	Each decision is attached to its Langfuse trace	Not built. The decision and its timestamp are in the n8n execution
FR-H6	Edited outputs are stored beside the original as correction pairs for later evaluation and fine-tuning	Not built. Future work
FR-H7	Clarify responses skip approval	Met
FR-H8	Anything addressed to the client always requires approval	Met. Every non-clarify output is approved

6. Inputs and outputs

6.1 Inputs

File	Fields and content
project_description.txt	Free text: ABCDE Ltd., logistics and supply chain. Goals of 15% fewer delivery delays, 10% lower operational costs and better service responsiveness. Focus on demand forecasting, route optimization, inventory management and customer support
project_timeline.csv	phase, phase_name, start_week, end_week, key_activities, milestones_deliverables (8 phases, 24 weeks)
project_risks.csv	risk_id, category, risk_challenge, impact, mitigation_strategy (10 risks, R01 to R10)
sample_task_board.csv	task_id, task_name, status, assignee, priority, sprint, due_date, labels, description (25 tasks, T001 to T025)
eval_mira_inputs.txt	The 12 baseline test inputs

6.2 Output formats

Markdown, converted to HTML for the approved email. Underscores in file names are kept by the "Middle Word Underscores" option on the To HTML node (KL-10).

6.3 Output templates

- **Project plan:** summary, goals, phases table (phase, activities, milestones, deliverables, dates), timeline overview, assumptions and gaps, data sources.
- **Risk assessment:** summary, risk matrix (ID, category, risk, impact, mitigation, source), top risks and rationale.
- **Weekly status report:** as-of date and sprint, overall health and why, counts by status, completed, in-progress and blocked tasks, upcoming milestones, risks and actions.

7. Evaluation and results

7.1 Metrics

Metric	Target	Result
Grounding and hallucination rate	0 invented tasks, risks or milestones	0 in Runs 2 to 5
Correct clarification rate	4 of 4 (T2, T4, T6, T9)	4 of 4
Factual accuracy (counts, blocked task)	100%	100% (T8, T10)
Risk relevance	Risks come from the register	Code checks plus an LLM judge. The judge agrees with a human 58% to 68% of the time (KL-8)
Cost per output	Under 1 USD	About $0.00032 per request, about $0.0006 for a plan
Latency	Under 60 seconds	Average model-call time 2.8 to 3.5 seconds. End to end about 10 to 12 seconds in Langfuse

7.2 Runs

Run	Change	Passed	Hallucinations	Avg latency (ms, 8 model-call tests)	Cost per request
Run 2	After orchestrator fix	12 of 12 (T11 was a false pass)	0	3,124	$0.000319
Run 3	T11 prompt fix and stricter check	12 of 12	0	2,888	$0.000318
Run 4	Risk-relevance judge added (T3, T7)	12 of 12	0	2,814	$0.000315
Run 5	Judge rule 8 (risks must be in the register)	12 of 12	0	3,472	$0.00032

The baseline runs from the Mira Eval workflow into the Mira Eval Results Google Sheet. Langfuse holds the traces, cost and the risk_relevance evaluator scores. The judge was calibrated against 33 human-rated answers (see the failure analysis).

7.3 Baseline test mapping

Test	Type	Pass condition	Result
T1	Plan (detailed)	All 8 phases with the right week ranges and milestones. Goals limited to the description	Pass. Intermittent wrong start date in Run 3 (KL-1)
T2	Plan (vague)	Clarify asks for scope, timeline and team size. No plan	Pass
T3	Risk (detailed)	Categorized risks from R01 to R10 with matching impact and mitigation	Pass
T4	Risk (vague)	Clarify asks for project details. No risks invented	Pass
T5	Status (data)	Sprint 3 tasks T007, T008, T009, T025 with correct status. Health Amber	Pass
T6	Status (no data)	States no task data is available	Pass
T7	Risk analysis	Exactly 3 risks, all from R01 to R10, each with a reason	Pass
T8	Tracking	T024 Blocked, T006 on the watch list, T008 at risk. Nothing overdue	Pass. T006 is Watch, not At risk (KL-3)
T9	Plan (edge)	Clarify asks for scope, goals and deliverables	Pass
T10	Status summary	Done 5, In Progress 3, To Do 16, Blocked 1, total 25	Pass
T11	Timeline	Phase 2 deliverables due 2026-04-21. Phase 3 deliverable due 2026-05-05 may be named as next	Pass from Run 3 (Run 2 false pass, KL-4)

Test	Type	Pass condition	Result
T12	Comms	Names T004, T005 and T006. No Sprint 3+ tasks. Professional tone	Pass. Intermittent "no blocked items" overclaim in Run 3 (KL-2)

8. Non-functional requirements

Area	Requirement	Status
Reliability	Same input gives materially the same facts (low temperature, code-computed numbers)	Met for counts and dates. Prose varies and has two intermittent errors (KL-1, KL-2)
Performance	Response under 60 seconds	Met
Cost	Under 1 USD per output	Met
Security	API keys in n8n credentials, never in prompts or logs. No client data in public repositories	Met. Workflow exports were checked for secrets before upload
Privacy	Send only the fields needed. Use sample data in public artifacts	Met. Anonymization before sending to the provider is still an open question
Maintainability	Workflows split into sub-workflows. Prompts versioned in Langfuse	Partly met. Model calls share one sub-workflow. Prompts live in the workflow nodes, not in Langfuse
Transparency	Every output lists its data sources and as-of date	Met

9. Prompting guidelines

- Give each agent a narrow role and the exact template to follow.
- Pass source data in labelled blocks and tell the model to use only those blocks.
- Require source IDs for every task, risk and milestone mentioned.
- State refusal behavior explicitly: if required data is missing, output the clarification.
- Use low temperature. Ask the orchestrator for JSON and validate it.

- Do not paste example outputs with real-looking project facts, because examples can leak into answers.

10. Release plan (as delivered)

Release	Scope	Exit criteria	Status
MVP	Data loader, router, clarify, status reporter, approval gate, Langfuse tracing	T5, T6, T10 pass	Done
v0.5	Planner, risk assessor	T1 to T4, T7 pass	Done
v0.8	Milestone tracker, sprint update	T8, T11, T12 pass	Done. Audience-tailored updates not built
v1.0	Full baseline run, cost analysis, documentation	12 of 12 pass, deliverables packaged	Done (Runs 2 to 5)
Post-v1	Task board API, pilot with a real project	PM feedback and time-saved measurement	Not started

11. Known limitations and future work

Ten limitations are logged as KL-1 to KL-10 in the Known Limitations tab of the results sheet and in the failure analysis. Future work: a code check of task and risk IDs against the board and register, a judge with the full task board, a second-model comparison, fine-tuning on corrected examples, rejection categories and regeneration, attaching decisions to Langfuse traces, correction pairs, and audience-tailored stakeholder updates.

12. Deliverables checklist

- Exported n8n workflow JSON and a GitHub README
- Architecture diagram
- Screenshots: workflow canvases, Mira in action, Langfuse traces and scores
- Baseline results for all 12 tests with failure analysis
- Cost per plan and evaluation observations

13. Risks and open questions

Risks: invented content despite guardrails, over-refusal on valid requests, mismatch between assumed and real CSV schemas, a judge that agrees with a human only about two times in three.

Open questions

1. Is the test as-of date of 2026-04-20 acceptable to the sponsor? Project dates run 2026-04-07 to 2026-08-04 (about 17 weeks) while the timeline describes 24 weeks.
2. Should overall health follow a fixed rule (any Blocked or overdue task makes it at least Amber) or model judgment? T5 expects Amber.
3. Is anonymization of client data required before it is sent to the model provider? Owner: Phina Enemuo.
4. Which export channel next: Google Docs or Slack? Email is built.

Decided: email provider is Gmail. Model provider is OpenAI only (gpt-4o-mini).